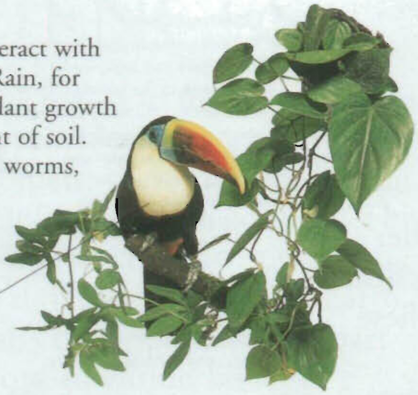
E

Ecological interactions

The components of an ecosystem interact with each other in lots of different ways. Rain, for example, provides water for plants. Plant growth and decay affect the form and content of soil. Soil provides a home for worms, and worms, as they move about, change the structure of the soil.

Puss moth larva cuts and chews leaves, using its sharp jaws.



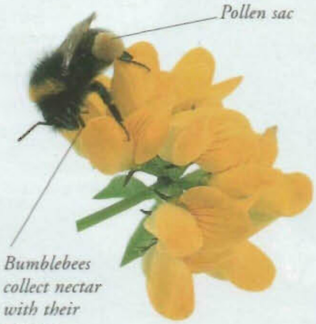
Toucans live high in the crowns of trees.

Food

Perhaps the most obvious way in which living species affect one another's lives is by feeding. Most things are food for something else. For example, caterpillars eat leaves, but are themselves food for animals such as birds. The birds are food for other animals, and so on up the food chain.

Shelter

The cover and shelter that trees and vegetation provide offer much more security than bare, open ground. In a rainforest, the large trees provide toucans with shelter from the weather, a place where they can raise their young in relative safety, and protection from predators.



Pollen sac

Bumblebees collect nectar with their tongue.

Transport

Animals can move around whereas plants cannot. Plants, therefore, use various methods that ensure animals carry their seeds and pollen, so that a new generation of plants can develop and grow. Bumblebees carry pollen on their legs.

Honey fungus

Young stinkhorn fungus

Parasitism

Animals, plants, and fungi that live off other living things are called parasites. Nearly all animals and plants are host to parasites of some kind. A parasitic relationship exists between a honey fungus and a tree. The fungus steals food from the tree, usually harming it in the process.

Symbiosis

When two species have a close relationship in which both benefit, it is called symbiotic. Symbiosis often involves giving shelter in return for protection or food, and it occurs among all kinds of organisms.

Clownfish

Clownfish find shelter among the stinging tentacles of sea anemones, which do not harm them. The fish may lure in other fish for the anemones to consume.

Clownfish stay where they are protected.

Adaptation

All plants and animals are specially suited to live in their particular habitat. How they become suited, or adapted, is the key to evolution. How and where a species lives, how it gets its food, what it eats, and how it interacts with others, is known as its ecological niche.

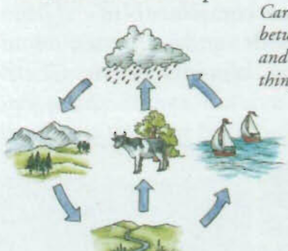
Spines protect the swollen stem.

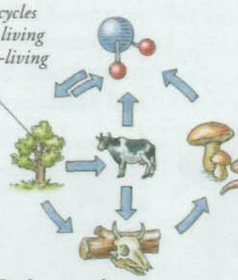
Cacti

A cactus has adapted in many ways to desert life. For example, its leaves have adapted into spines, to prevent water from evaporating too easily. When rain does fall, a cactus stores as much water as possible in its stem.

Cycles in nature

Nature automatically recycles the substances that are vital for life. Oxygen, nitrogen, carbon, and water are constantly being exchanged between the air, the soil, the oceans, and living things. If substances were not continuously put back into the ecosystems to be used again, the supply for organisms would soon run out and life would stop.





Carbon cycles between living and non-living things

Water cycle

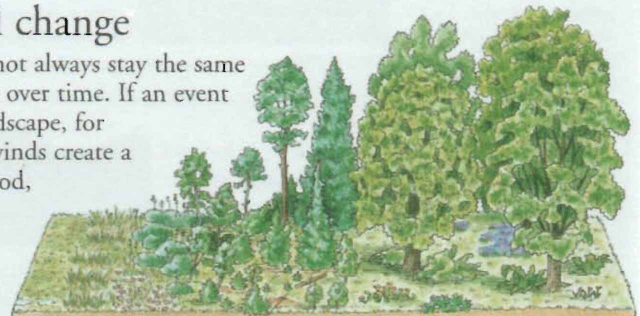
Water lost by evaporation from plants, rivers, and seas, forms clouds in the atmosphere. This falls back as rain, runs into rivers and seas, and is soaked up from the soil by the roots of plants.

Carbon cycle

Organisms release carbon dioxide into the air. Carbon is also released when organisms decay, or when coal is burned. Plants absorb carbon from the air, which passes into animals that eat them.

Ecological change

Ecosystems do not always stay the same but may change over time. If an event changes the landscape, for example, high winds create a clearing in a wood, first grasses and herbs grow, then shrubs colonize the plot until trees take over once again.



The process of change from grassland to woodland is called succession.

Land erosion in Madagascar

Human impact

People's actions also change ecosystems and often the impact is so great that nature cannot repair the damage. For example, poor farming techniques sometimes cause so much soil to be eroded away from the land, that plants cannot get established and the vegetation can never recover.

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